

Contributions of GPS Permanent Station and Seismological Observatory to Geophysical Studies at Maitri, Antarctica During XXIV-Indian Antarctic Expedition

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ABSTRACT

On going research activities at Antarctica were successfully carried out during XXIV IAE for the long-term scientific objectives of NGRI. The data from GPS station continues to contribute to Scientific Committee on Antarctic Research (SCAR) and International Earth Rotation Service (IERS). The seismological observatory continues to contribute to NEIC (USGS), ISC, (UK) and plays a pivotal role in the Global Seismographic Network in Antarctica. Global positioning system (GPS) measurements at Maitri, Indian Antarctic station, have been carried out for the period 1997 to date. The International GPS Service for Geodynamics (IGS) stations, in different plates, such as Casey, Davis, Kerguelen, Seychelles, Coco, Hartebeesthoek, Yaragadee, and Tidbinbilla were chosen for global network solution. The GPS data from 1997 to 2005 of all these IGS stations in and around Antarctica along with Maitri were used to estimate the time series, baseline lengths and their changes and velocity vectors of individual sites. The results have enabled to obtain a deeper insight into the driving mechanism and the causative factors for Indian plate kinematics. During this expedition i.e., from January 2005 to February 2006, continuous seismic data were also acquired for processing and analysis. The final analysis of this data has been completed using SEISAN software and reported to ISC, United Kingdom as per their time schedule up to December 2005. The number of earthquakes recorded as global events and near-events in and around Antarctica during the period of expedition is discussed in this report.

INTRODUCTION

Focused study on the Indian Ocean and South Sandwich Islands Region becomes very important in the present scenario of interplate and intraplate earthquakes that do occur frequently in India and nearby regions,

forcing us to re-evaluate the rigidity of the Indian plate as well as Indian Ocean Margin. GPS-Geodesy and Seismic Studies are extended to envelop the large Indian Ocean lithosphere. In order to monitor the seismic activity in and around Antarctica and Indian Ocean, a reconnaissance survey for site selection and feasibility of operation of seismological observatory in Antarctica was initiated during 16th Indian Antarctic Expedition and a Digital Short Period System was installed during 17th expedition. The station was fully commissioned on 26 January 1998 with analog and short period digital systems. This observatory was further upgraded with the installation of Broadband seismic system during the 20th IAE, comprising a Broadband Sensor CMG-3ESP (GURALP make with 50 Hz to 100 sec flat response) and Reftek Data Acquisition System.

The permanent seismic station at Maitri has gone global by figuring in AnSWeR. (Antarctic Seismic Web Resource) as shown in Figure 4, with the station code MAIT. After final analysis of seismic broad band digital data, the complete phase data had been reported to ISC, to be included in the global data center for global access by any other agency and can be accessed at: <http://www.isc.ac.uk>

The Seismic sensors (S-13 for short period & analog recorders and Guralp CMG-3ESP for Broadband System) have been installed in the underground vault behind Tirumala Hut. The Data Acquisition System (DAS) – REFTEK High Resolution Digital Data Acquisition System RT 72A 121-03 from Refraction Technology Inc., USA interfaced with a P.C., Analog Recorder – vertical component analog RV-320B Portacorder and a GPS receiver – was installed in the Tirumala hut. The Sensor pit and the lab are being maintained at about 15 to 20 degrees temperature using electrical heaters. The data acquired have been processed using SEISAN software. During this expedition i.e., from January 2005 to February 2006, continuous seismic data have been recorded and archived and stored in CDs for further processing and analysis.

Using SEISAN software the final analysis of this data collected up to December 2005 has been completed and reported to International Seismological Centre, United Kingdom. The data acquisition during wintering period, the Preliminary Determination of Epicenters (PDE) reports by NEIC are referred to while processing the data using SEISAN 7.2, PCSUDS and REFTEK utilities software. After completion of the analysis, the arrival times, origin times and the corresponding hypo central parameters of the processed events were used to compile the MAITRI Seismological Bulletin for the year 2005.

The India–Eurasia collision zone has been the focus of intense studies by the Space Geodetic Community for Indian Plate Motion. So far space geodetic studies by GPS have been made in southern and central India that bears on the rigidity of the Indian plate, whose velocity data for two sites fit the rigid plate model within uncertainties, estimated at about 3.7 cm/yr. for Hyderabad and 3.9 cm/yr. for Bangalore, respectively. Since Space geodesy can test many otherwise untestable predictions and boundary conditions, GPS-Geodesy has been extended to envelop the large Indian Ocean lithosphere and observations have been extended to the Indian Ocean and Antarctica. Despite the extensive studies to date, major issues remain unresolved. Motions across some of the plate boundaries seem well constrained, as implied by good agreement between space geodetic and geologic models. In others, apparent discrepancies exist. Very few studies have been conducted on the larger oceanic part of the Indian plate using space geodesy. To holistically determine the kinematics of the Indian Ocean Basin between Antarctica and India, the data available are very sparse and characterization and delineation of the plate boundaries especially in the Indian Ocean are poor. This study addresses several of these issues mentioned as follows :

- (1) How rigid is the Indian Plate and Indian Ocean Basin?
- (2) Does relatively high level of intraplate seismicity on the oceanic part of the Indian Plate indicate internal deformation in excess of other plates? and,
- (3) Is this related to Indo-Eurasian collision and the uplift of Himalayas?

Therefore, GPS geodesy is used to improve the understanding of the complex plate motions, diffuse and poorly located plate boundaries, and striking intraplate deformation that characterize the Indian Ocean basin.

Work Carried out During the Expedition

Acquisition of : A) GPS Data of Permanent GPS Station at Maitri, Antarctica.

and

B) Seismological Data at Maitri, (in Tirumala Hut), Antarctica.

A) GPS Data of Permanent GPS Station at Maitri

During 24th IAE i.e., from January 2005 to February 2006 the following works were carried out by Shri A. Nageswara Rao (NGRI) XXIV-IAE Winter Member:

- 1) Installed Turborogue GPS system in place of Tremble GPS receiver.
- 2) Operated continuously and acquired GPS data of both Ashtech and Turborogue during the period January 2005 to February 2006
- 3) The daily data is of the size of 3.2 Mega bytes (approximately), with 15 sec sample rate.
- 4) Pentium III computer was used for downloading the data via “HYPER” terminal software. The receiver acquired the data continuously which was converted into RINEX files. The acquisition and pre-processing of data were done regularly with maximum care.
- 5) The data were brought back to NGRI for further processing by using Bernese 4.2 version and analysis. The Global network of GPS stations included Maitri (MAIT), Casey (CAS1) and Davis



Fig. 1 : GPS Antenna at Antarctica

(DAV1) in the Antarctic plate, Seychelles (SEY1) in Somalian plate, Cocos (COCO) on edge of Indo-Australia diffuse plate boundary, Yaragadee (YAR1), Tidbinbilla (TID2) and Australian plate and Hartebeesthoek (HRAO) in African plate in our data processing.

6) The GPS data was contributed to SCAR.

B) Acquisition of Seismological Data at Maitri, (in Tirumala Hut), Antarctica

- 1) Operated One Single Component Seismograph (Analog-Portacorder RV 320B) and one Three Component Broadband Seismograph (Digital) by Sri A. Nageswara Rao (NGRI) as a Winter Member of XXIV-IAE. The two seismographs were operated continuously and acquired seismic data of both analog and digital during the expedition period January 2005 to February 2006. The Seismological Bulletin was prepared for the period January 2005 to December 2005 (Technical Report No. NGRI-2008-ANT-617) and submitted to the concerned authorities.
- 2) To improve the system performance we have opened the seismic vault and completed the sensor leveling and mass centering by locking and unlocking. The entire seismic system was

TIRUMALA OBSERVATORY (NGRI), MAITRI,
ANTARCTICA, SEISMOGRAPHS: ANALOG & DIGITAL



*Fig. 2 : Seismological Observatory (NGRI) at
(in Tirumala Hut) Maitri, Antarctica*

refurbished for satisfactory operation. After doing this, we observed that the daily file size improved from 20 to 25 MB (as it was earlier). There was improvement in the signal level also.

- 3) Temperature was maintained properly in the observatory and seismic vault. The temperature was maintained between 20° and 25°C with the help of the temperature sensor and thermometer.

RESULTS AND DISCUSSION

- A) **GPS Data:** During XXIV-Indian Antarctic Expedition, uninterrupted good quality of GPS data of both Ashtech and Turborogue was acquired and processed at NGRI for further analysis.
- B) **Seismic Data and Results:** Uninterrupted good quality Seismic analog & Digital Broad Band data were acquired. The data were processed and analyzed at NGRI. For the Digital Broad Band seismic data, SEISAN software was used for the final processing and analysis. In the Indian Antarctic Station, Maitri, a permanent Seismological Station was commissioned in 1998 with an analog recorder. Subsequently, it was upgraded with a Broad Band Digital System in January, 2001. The Broad Band Seismic data were contributed to International Seismological Center (ISC), U.K. The Processing of the data up to 2005 has been completed and the seismological Bulletin of Maitri Station for 2005 has been published.

The total number of earthquakes recorded from January to December 2005 were 329. **Figure 3** shows that the epicentral map of the events recorded during 2005.

An earthquake of magnitude of 8.4 Ms was recorded on 28th March, 2005 (**Fig. 8**) at a distance of about $\Delta=90.40$ located at Northern Sumatra (2.074 N; 97.013 E; Depth = 30 km). Three earthquakes of magnitude > 7 Ms were recorded during the period from January to December, 2005. They are: 1) on 24th July, 2005 of magnitude 7.5 Ms (**Fig. 7**) at a distance of $\Delta=94.4$ located at Nicobar Islands, India (7.909 N; 92.139 E; Depth= 16 km); 2) on 8th October, 2005 of magnitude 7.7 Ms (**Fig. 11**) at a distance of $\Delta= 114.0$ and located at Pakistan (34.476 N; 73.577 E; Depth = 26 km) and 3) on 5th December, 2005 of magnitude 7.2 Ms at a distance of $\Delta= 65.6$ and the location was Lake Tanganyika Region, South Africa (6.174 S; 29.717 E).

Apart from the above, two nearest earthquakes were also recorded. They are : 1) from Bouvet Islands region of Mb 5.3 (53.966S; 7.760 E) on

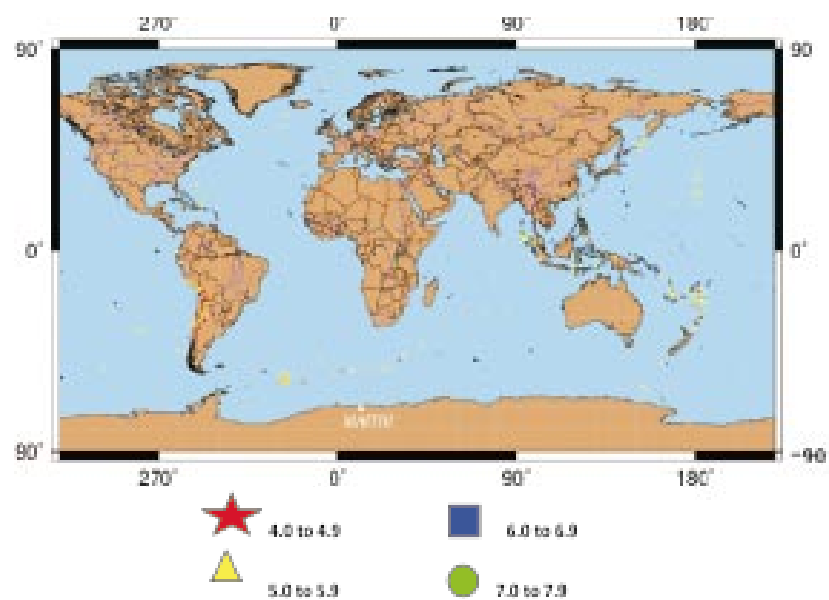


Fig. 3 : Epimap of 329 earthquakes recorded by NGRI Seismic Station at Maitri, Antarctica

Maitri in Antarctic Seismic Web Resource (AnSWeR)

Antarctic Broadband Seismic Observatories

Filled triangles=Existing stations, Open triangles=Proposed stations

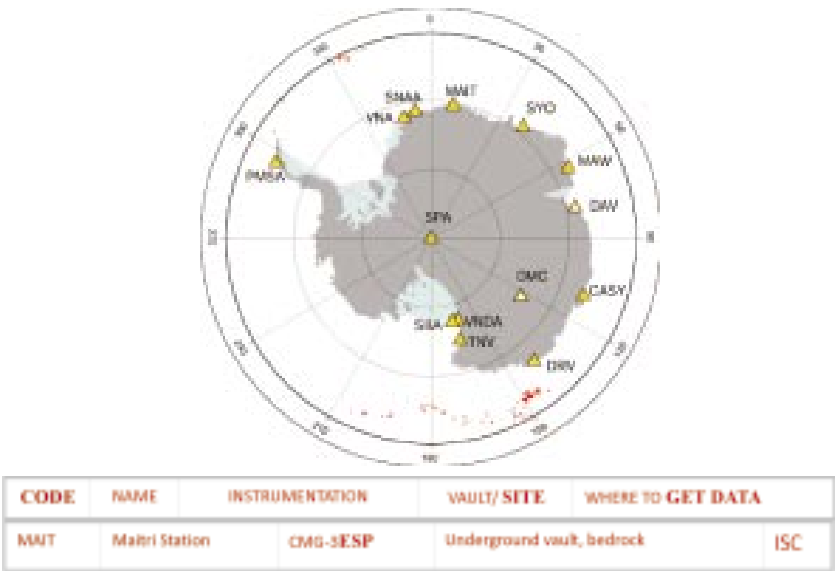


Fig. 4 : Permanent Seismological Observatory in AnSWeR

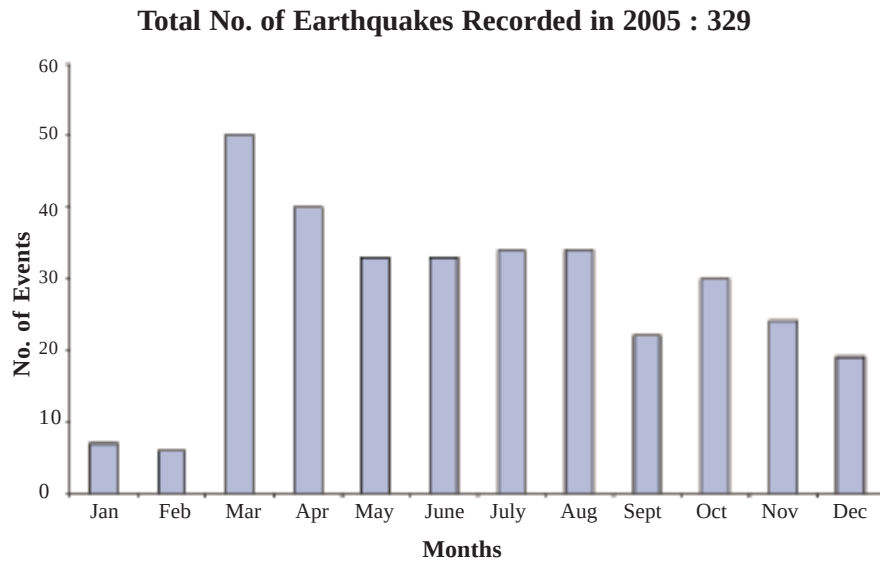


Fig. 5 : Monthly distribution of earthquakes recorded at Maitri Seismological Observatory (MAIT), Antarctica in 2005

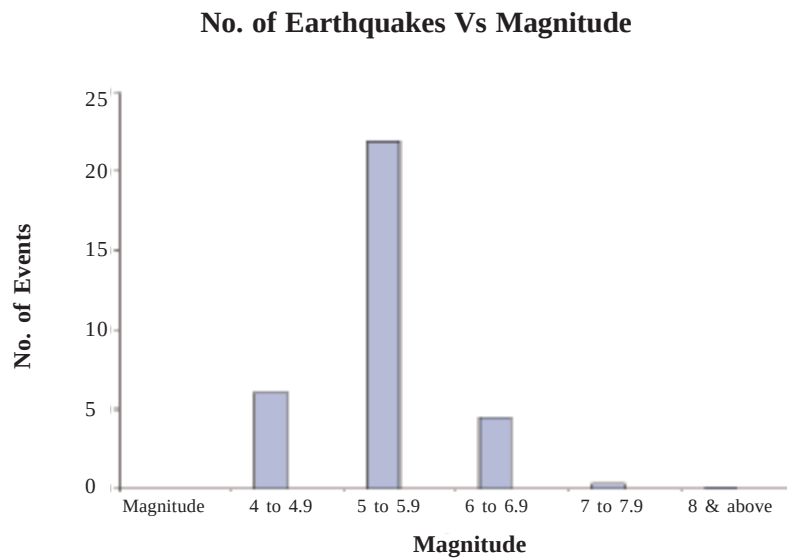


Fig. 6 : Number of earthquakes recorded in 2005 vs magnitude

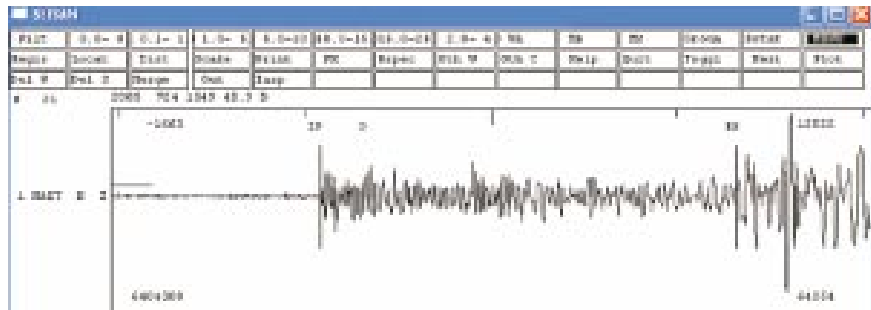


Fig. 7 : Nicobar Islands Earthquake of Ms 7.5 (7.909 N; 92.139E) at a distance of Δ 94.4° and depth 16km recorded by NGRI-Seismograph at Maitri, Antarctica on 24-07-2005

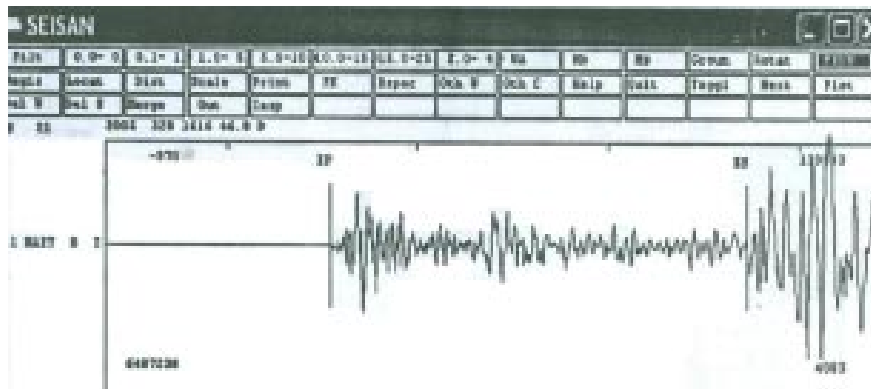


Fig. 8 : Northern Sumatra Earthquake of Ms 8.4 (2.074 N; 97.013 E) was recorded on 28th March, 2005 at a distance of Δ 90.4° and depth 30 km. Earthquake recorded by NGRI Seismograph at Maitri, Antarctica

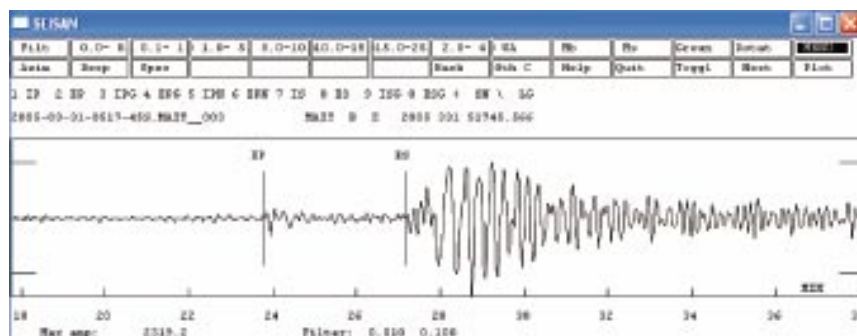


Fig. 9 : Nearest earthquake of Bouvet Islands region of Mb 5.3 (53.966 S; 7.760 E) on 31st March, 2005 at a distance of Δ 16.9° and depth 10 km recorded by NGRI Seismograph at Maitri, Antarctica

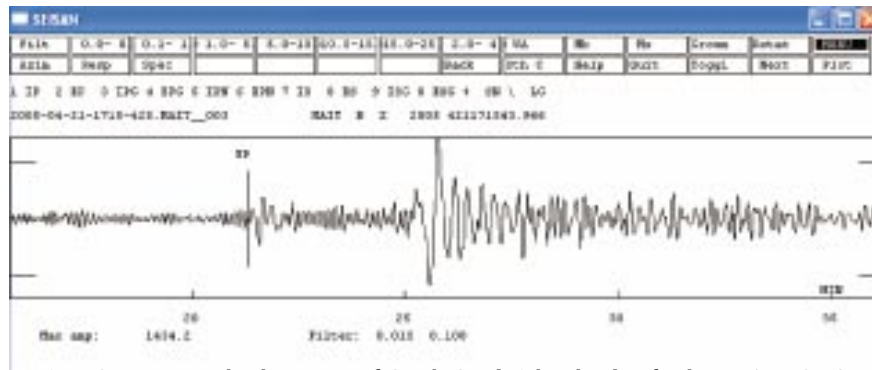


Fig. 10 : Nearest shock at east of South Sandwich Islands of Mb 4.9 (59.109 S; 16.975 W) on 21st April, 2005 at a distance of Δ 16.5° and depth 10 km was recorded by NGRI Seismograph at Maitri, Antarctica

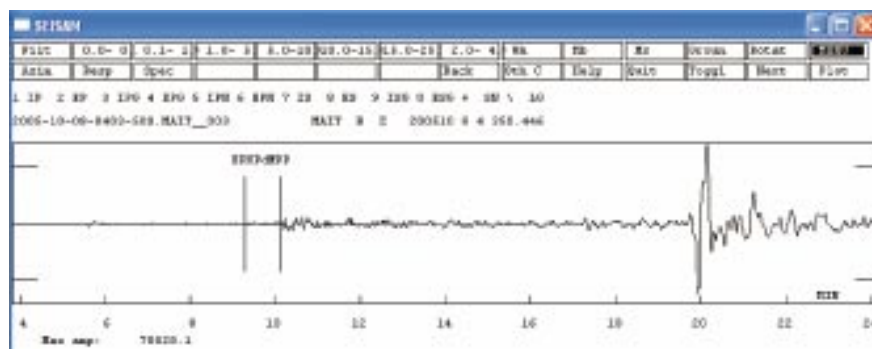


Fig. 11 : Pakistan Earthquake of Ms 7.7 on 8th October, 2005 (34.476 N; 73.577 E) at a distance of Δ 114° and depth 26 km recorded by NGRI Seismograph at Maitri

31st March, 2005 (**Fig. 9**) at a distance of “ 16.9° and depth 10 km recorded by NGRI Seismograph at Maitri, Antarctica and 2) from South Sandwich Islands region of Mb 4.9 (59.109 S; 16.975 W) on 21st April, 2005 (**Fig. 10**) at a distance of Δ 16.5° and depth 10 km.

CONCLUSIONS

The Seismic data analysis and results show the strain accumulation and deformation processes towards the Indian Plate. These studies become all the more important in view of the recent activities after the Sumatra earthquake and subsequent events. Therefore these studies need to continue for a longer span of time to precisely estimate the seismicity and tectonic activity in and around Antarctica and in the Indian Ocean. The collocation

of the permanent GPS Station and Seismic Observatory will also result in monitoring the space and time distribution of earthquake occurrences and obtain hypo-central parameters, magnitudes of earthquakes, release of energy, Strain accumulation and stress drop, velocity inversion for underground structure and physical properties, earthquake source mechanism, receiver function analysis, attenuation of seismic waves and anisotropy studies.

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